

The Blank Operating System

Team Vertex (Group 23)

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Abstract

The Blank Operating System Layer is a user interface that makes operating system navigation and control more expressive and intuitive. It consists of a large language model, a single client, and multiple servers that communicate through the Model Context Protocol, with each message written in JavaScript Object Notation format.

The large language model can be provided by an existing system such as ChatGPT, Google Gemini, or a self-hosted local model.

Each server is responsible for a specific group of related functions. For example, one server may handle reading and writing to the file system, while another may provide weather information from the internet. Every server exposes the purpose of its available functions, along with parameter types and return values, in a format that follows the Model Context Protocol standard. This design restricts the large language model to these specific functions, and any invalid function calls or parameters are rejected by the server.

The client provides a terminal-style interface that allows users to issue tasks in natural language through typing in the text entry or using the speech-to-text feature. When a task is submitted, the client forwards it to the large language model, which interprets the user's intent and converts it into a single Model Context Protocol message. If the task cannot be performed based on the available servers, the large language model rejects it and returns an error. If the task is ambiguous, such as creating a temporary file, the large language model may ask for additional details, including the file's name, type, and lifetime in a conversation dialog. Once the client receives the message from the large language model, it displays the task details, including the operation to perform, generated parameters, required permissions, and server dependencies, on a dialog, for the user to review. The user can type yes to continue or no to cancel. If the user chooses to continue, the client sends the message to the appropriate server, which executes the task.

Third-party applications with Model Context Protocol library installed, can interact with the client through its exposed application interface. For instance, it may directly send a task to the client as if the user types in the text entry.

Nouns

The Blank OSL: The Blank Operating System Layer.

JSON: JavaScript Object Notation

API: Application Programming Interface.

LLM: Large Language Model.

UI: User interface.

AI: Artificial intelligence.

OS: Operating system, but restricted to only Windows in The Blank OSL for the sake of simplicity.

MCP: Model Context Protocol, an open-source standard for connecting AI applications to external systems.

Natural Language: Human speaking language, but restricted to only English in The Blank OSL for the sake of simplicity.

Task: An action described by natural language as a paragraph, for example, “getting the titles of the top 10 most popular songs in 2025, then save to my home directory as hello.txt”.

Message: A JSON encoded message following the MCP format.

Client: A client that implements MCP client API.

Server: A server that implements MCP server API.

User: Human user who uses The Blank OSL.

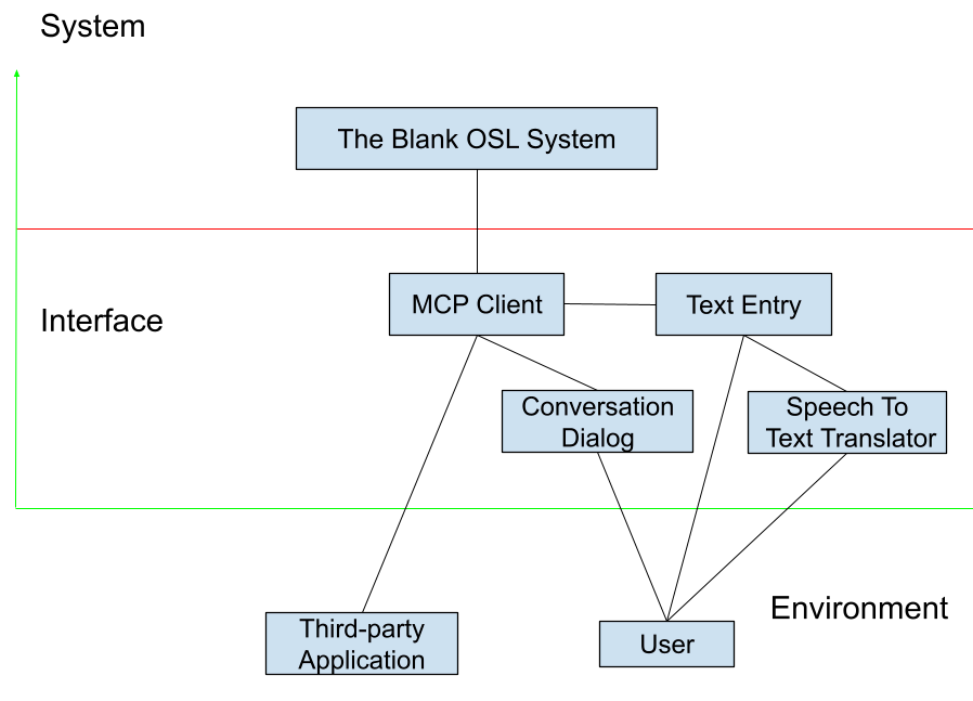
Third-party application: Any external software with MCP support that can communicate with the Blank OSL client via its API.

Dialog: A conversation dialog that contains either error message, or MCP messages with details and ask for user’s confirmation.

Domain Model

The Blank OSL

Group 23



The LLM is in the system domain as the user does not directly interact with it. In fact, it is completely controlled by the MCP library and abstracted away under the MCP client.

The MCP servers are in the system domain as it contains most of the function implementation details. MCP client communicates with multiple MCP servers through the Model Context Protocols.

Third-party applications may directly interact with the MCP client if they follow the MCP format.

Use Case Model

Actors

User: Interact with the client through text entry, voice to speech translator, and the dialog.

Third-party Applications: Interact with the client through the client API.

Use Cases

UC1: Submit Task via text or speech.

UC2: Review the message information in the conversation dialog.

UC3: Approve or disapprove the task action in the conversation dialog.

UC4: Provide additional context in the conversation dialog.

UC5: Acknowledge the error in the task action in the conversation dialog.

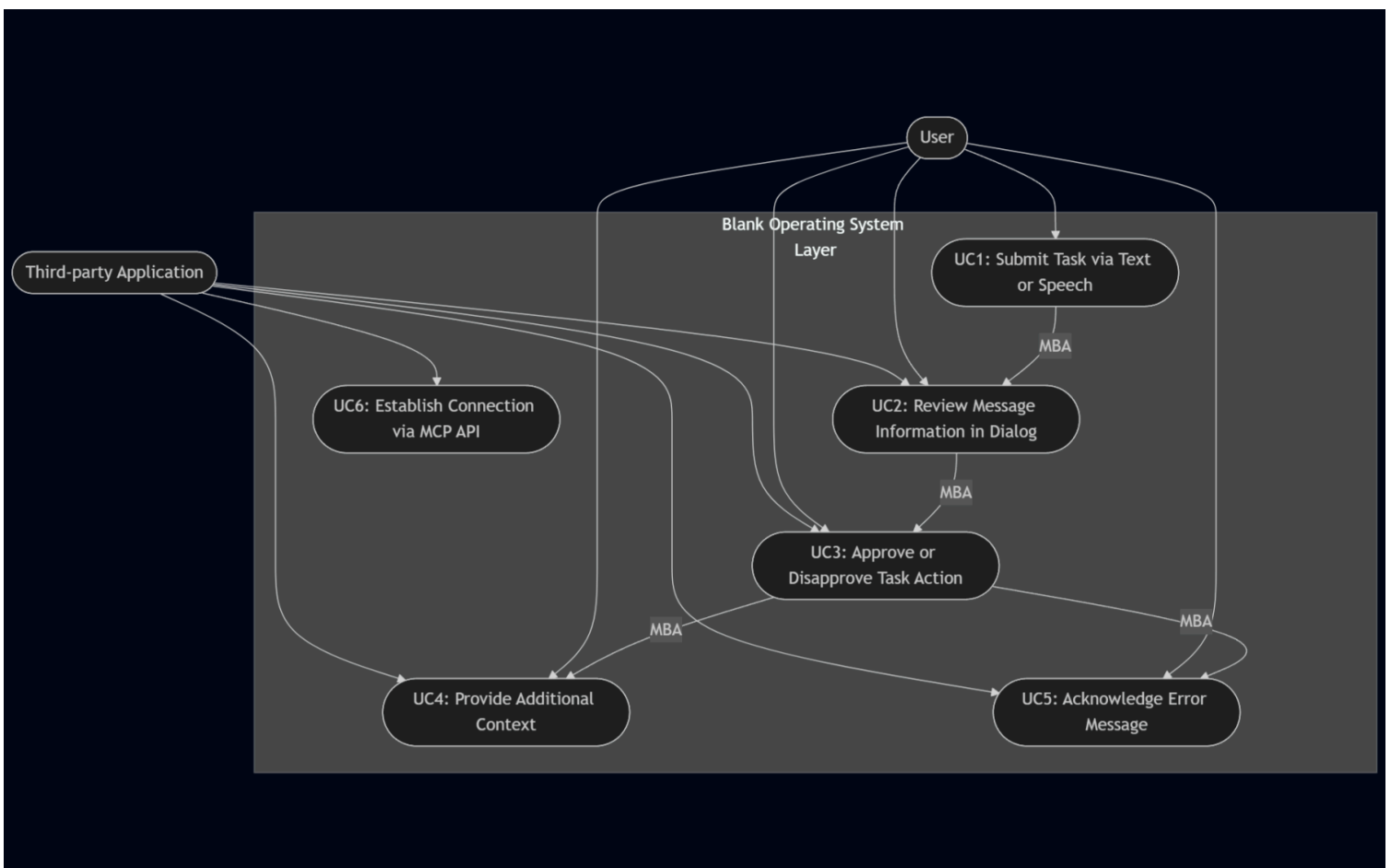
UC6: Connect third-party applications to the client via MCP API.

Use Case Mapping

User through client interface: UC1-UC5

Third-party Application interact programmatically: UC1-6

Use Case Diagram



Domain Assumptions

1. The Blank OSL runs on a Windows 10 or Windows 11 operating system.
2. Only one instance of the Blank OSL client is running at a given time.
3. All Blank OSL components (client, servers, and LLM) function normally.
4. The LLM used has sufficient token capacity for all tasks.
5. The speech-to-text translator performs reliably regardless of the user's accent.
6. The Blank OSL client and servers can finish the task in a reasonable time frame.
7. The user has a computer with basic input hardware such as keyboard and microphone.
8. The user can type, read, and speak English.
9. The task that the user provides matches what they actually want to do.
10. The user understands basic Windows navigation, e.g. they can navigate the UI.
11. The user and third-party applications do not attempt malicious actions.
12. The third-party application must comply with the Model Context Protocol.

Variations

1. Submit a task via text entry.
2. Submit a task via speech-to-text.
3. Provide additional context through the dialog to complete multi-step tasks.

Exceptions

1. The task length exceeds LLM token limits
2. The task is described in non-supported natural languages or gibberish like "el ess asterisk".
3. Task execution is manually terminated.
4. The task is not supported by any of the MCP servers available.
5. The task is empty.
6. The task requires external resources that are not available.
7. The task is malicious and harms the computer.
8. The task has minor grammatical or spelling errors.
9. The microphone fails to pick up the user's voice.
10. The computer loses power and shuts down.